

1. Questions

Read the given information carefully and answer the following questions based on it.

The missing table given below shows the total number of notes of three denominations of Rs 20, Rs 50, and Rs 200, the ratio of the number of notes of Rs 50 and Rs 200, the difference between the number of notes of Rs 20 and Rs 200 and the total amount (in Rs) of four persons. Some values are missing from the table.

Name	The total number of notes	The ratio of the number of notes of Rs 50 and Rs 200	The difference between the number of Rs 20 and Rs 200 notes	Total amount (in Rs)
Karan	90	-	10	-
Saroj	-	5:3	21	3150
Harish	-	6:1	-	2900
Jay	-	10:3	25	-

Notes:

- The number of notes of Rs 20 is more than that of Rs 200 for each.
- The ratio of the number of Rs 50 notes of Karan, Saroj, Harish and Jay is 8:3:6:10.

The ratio of the number of Rs 20 notes of Harish and Karan is a:b (minimum possible). If Saroj had 'a*b' more notes of Rs 10, then the total amount of Saroj is approximately what percentage of the total amount of Jay?

- 63%
- 51%
- 45%
- 71%
- None of these

2. Questions

Which of the given statements is or are false?

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- The difference between the amount of Karan and Jay is Rs 300.
 - The sum of total number of Rs 200 notes of all four persons is 49.
 - Harish's total number of notes is 45% of Karan's total number of notes.
- Only I
 - Only II
 - Only III

- d. Both II and III
- e. None of these

3. Questions

Establish the relationship between the given quantities. (consider numerical values only)

Q I: Difference between the total number of Rs 50 notes of Harish and Jay together and the total number of Rs 20 notes of Karan and Saroj together.

Q II: How much more or less is the average total number of notes of Jay and Harish than the total number of notes of Rs 200 of all four?

Q III: 20% of the difference between the amount of Saroj and Harish.

- a. $QI = QII > QIII$
- b. $QI > QII < QIII$
- c. $QI = QII = QIII$
- d. $QI < QII < QIII$
- e. None of these

4. Questions

Which of the following statements is or are sufficient to answer the given question?

What is the total number of notes that have deep?

I). Deep has total amount of Rs 6900. He has a total of 45 denominations of Rs 20.

II). Difference between the number of notes of Rs 20 and 200 of Deep is 30. Number of notes of Rs 50 of Deep is 20% more than that of Jay.

- a. Only I
- b. Only II
- c. Both I and II together
- d. Either I or II
- e. Neither I nor II alone

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5. Questions

The total number of notes of Mahesh is 8 more than the average total number of notes of Karan and Saroj. If Mahesh has 25 notes of Rs 50, then find the number of notes of Rs 20 when he has highest number of notes of Rs 200.

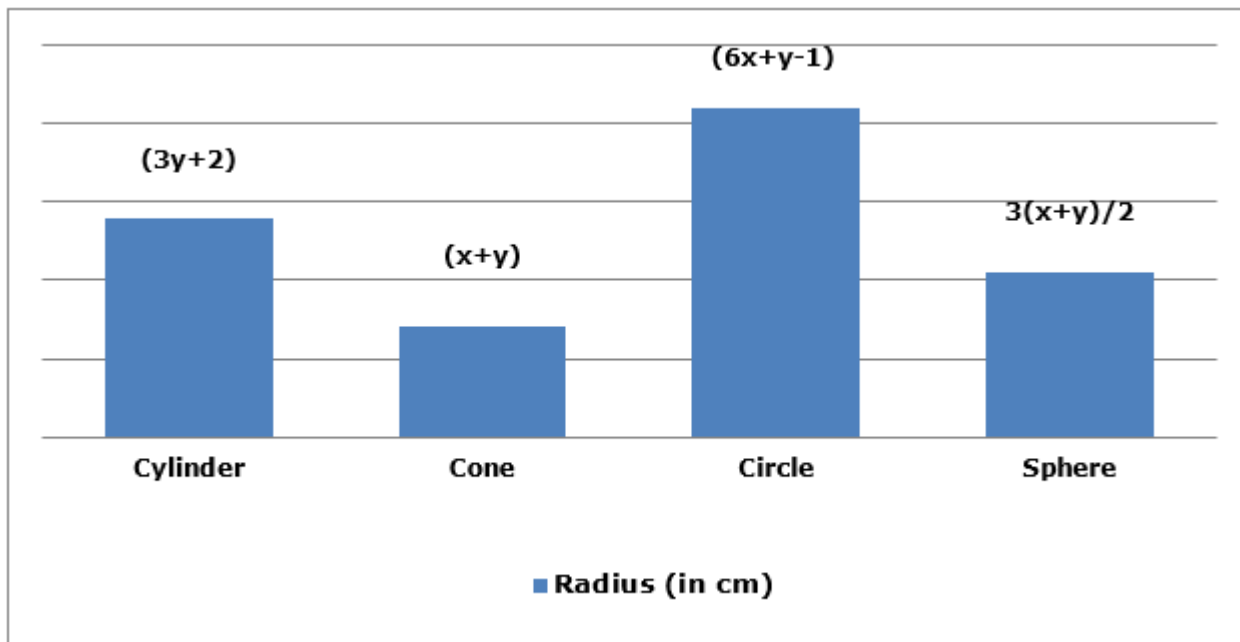
- a. 27
- b. 28

- c. 30
- d. 26
- e. None of these

6. Questions

Read the given information carefully and answer the following questions based on it.

The given bar graph shows the value of radius of cylinder, cone, circle and sphere respectively.



Notes:

- I). The perimeter of the circle is 132 cm. The volume of the sphere is 4851cm^3 .
- II). The slant height of cone is 25 cm. The height of cylinder is x cm more than that of cone.

Which of the given statements is or are true?

- I). The radius of the circle is 50% of the height of the cylinder.
- II). Curved surface area of the cone is 500cm^2
- III). The average radius of the cone and circle is 15 cm.

- a. All are true
- b. Only I
- c. Both II and III
- d. All are false
- e. None of these

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7. Questions

Establish the relationship between the given quantities:

Q I: Difference between the area of the circle and the base of the cylinder.

Q II: Total surface area of the sphere.

Q III: Curved surface area of the cylinder.

- a. $QI = QII > QIII$
- b. $QI > QII < QIII$
- c. $QI = QII = QIII$
- d. $QI < QII < QIII$
- e. None of these

8. Questions

The height and base of a cuboid is 6 cm more than the radius of the cylinder and 10 cm less than the height of the cone respectively. The length of cuboid is $3x$ cm. Find the total surface area of the cuboid.

- a. 1172 cm^2
- b. 1440 cm^2
- c. 1120 cm^2
- d. 1650 cm^2
- e. None of these

9. Questions

The given sphere is melted and formed into $(30y+n+1)$ number of small spherical marbles whose radius is 1.5 cm. What is the value of n/x ?

- a. 63
- b. 74
- c. 81
- d. 65
- e. None of these

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10. Questions

Breadth of a rectangle is same as the slant height of a cone and its Length is $3x$ cm more than the radius of circle. The perimeter of a square is same as the perimeter of circle. Find difference between area of rectangle and square.

- a. 430 cm^2
- b. 366 cm^2
- c. 390 cm^2
- d. 540 cm^2
- e. None of these

11. Questions

Read the given information carefully and answer the following questions based on it.

There are some sports persons who are using different brand out Puma, Addidas and Gucci. Each person likes at least one brand out of the three.

Total number of people who wear both Puma and Addidas is 38. Ratio of the number of people who wear only Puma and only Addidas is 11:40. Total number of persons using Gucci brand is 42.5% more than the number of people using only Addidas. The total number of people using Puma is 95. The number of people wearing both Gucci and Addidas but not Puma is 100 less than the total number of people wearing Gucci. A total of 132 people use the Addidas brand.

Total number of people who use both Gucci and Addidas is what percent of the total number of people who use only Puma and only Gucci together?

- a. 35%
- b. 52.5%
- c. 46.1%
- d. None of the given options
- e. Cannot be determined

12. Questions

The total number of people who use exactly two brands is approximately what percent of the total number of sport persons if number of people using exactly three brands is 10?

- a. 32%
- b. 44%
- c. 28%
- d. 40%
- e. None of these

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13. Questions

Which of the given statements are true?

I). Ratio of the number of people who use Addidas and Gucci but not Puma to the number of people who use Puma and Gucci but not Addidas is 2:5.

II). Total number of people using Puma is 80% of the total number of people using Gucci.

III). Average of number of people who use only Puma and only Addidas is 50.

- a. Only I
- b. Both I and II
- c. All three
- d. Both I and III
- e. None of these

14. Questions

Establish the relation between given quantities:

Q I: If the number of people who use both puma and Addidas but not Gucci is 100% more than the number of people who use both Gucci and Addidas but not Puma, the find the total number of people using exactly one brand.

Q II: Average of total number of people using Puma, Addidas and Gucci.

- a. $QI > QII$
- b. $QI < QII$
- c. $QI \geq QII$
- d. $QI \leq QII$
- e. $QI = QII$ or relation cannot be established

15. Questions

‘m’ is the difference between the number of people who use only Addidas and total number of people using Puma while ‘n’ is the sum of total number of people who use Puma only and the number of people who use both Addidas and Gucci but not Puma. (n-m) is the total number of people using exactly all three brands. What is the total number of sport persons?

- a. 244
- b. 256
- c. 233
- d. 296
- e. None of these

16. Questions

In each question a number series is given. Read the given information carefully and answer the

questions based on it after solving.

The given series contains a wrong number.

1500, B, 1556, 1340, 1852, A

$$A = 50\% \text{ of } B + 104$$

$$B = 36^2 + 200$$

What is the correct term of the given wrong number series?

- a. 1562
- b. 1845
- c. 850
- d. 1492
- e. None of these

17. Questions

One term of the given number series is missing.

4000, (Y+300), 2400, X, _____, 288

$$(3x+2y) = 2440$$

$$\text{And } (2x-y) = 460$$

There is a new series that follows the same pattern of the given series and its first term is 60 more than the missing term of the given missing number series. What will be the 3rd term of the new number series?

- a. 180
- b. 900
- c. 630
- d. 960
- e. None of these

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18. Questions

The given series is a missing number series.

Series 1: 10000, A, 12.5, (B÷2), _____

$$A/B = 100$$

$$A+10B = 275$$

Series 2: 10A, D, _____, C, E

Series 2 follows the same pattern as series 1. Find the sum of the missing terms in both series.

- a. 2.725
- b. 5.75
- c. 3.375
- d. 4.325
- e. None of these

19. Questions

The given series contains a wrong term.

Series 1: 1250, 1183, 1112, 1039, 1000

A new series follows the same pattern as the given series and its first term is the wrong term of series 1. What is the average of the 4th and 5th terms of the new series?

- a. 749.5
- b. 569.4
- c. 670
- d. 664.4
- e. None of these

20. Questions

The given series is a wrong number series.

Series 1: Z, X, 437, Y, 710, 966

$$(X+Y) = 920$$

$$(Y-Z) = 230$$

$$(X-Z) = 46$$

What is the difference between the correct term and the last term of the given series?

- a. 147
- b. 253
- c. 341
- d. 211
- e. None of these

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Explanations:

1. Questions

For Saroj

Let the number of notes of Rs 50 = $5x$

Number of notes of Rs 200 = $3x$

Number of notes of Rs 20 = $(3x+21)$

$$50*5x + 200*3x + 20*(3x+21) = 3150$$

$$250x + 600x + 60x + 420 = 3150$$

$$910x = 2730$$

$$x = 3$$

The number of Rs 50 notes of Saroj = $5 * 3 = 15$

The number of Rs 200 notes of Saroj = $3 * 3 = 9$

The number of Rs 20 notes of Saroj = $9 + 21 = 30$

The ratio of number of Rs 50 notes of Karan, Saroj, Harish and Jay = 8:3:6:10(40, 15, 30, 50) respectively

Let the number of Rs 20 notes of Harish = A

$$30*50 + 200*5 + A*20 = 2900$$

$$2500 + 20A = 2900$$

$$20A = 400$$

$$A = 20$$

Let

The number of Rs 200 notes of Karan = x

The number of Rs 20 notes of Karan = $10 + x$

$$10 + x + x + 40 = 90$$

$$2x = 40$$

$$x = 20$$

The number of Rs 200 notes of Karan = $x = 20$

The number of Rs 20 notes of Karan = $10 + x = 30$

The total amount of Karan = $30 * 20 + 40 * 50 + 20 * 200 = \text{Rs. } 6600$

Let

The number of Rs 200 notes of Jay = $50/10 * 3 = 15$

The number of Rs 20 notes of Jay = $15 + 25 = 40$

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The total amount of Jay = $40 * 20 + 50 * 50 + 15 * 200 = \text{Rs. } 6300$

Name	Total number of notes	Number of notes of Rs. 50	Number of notes of Rs. 200	Number of notes of Rs. 20	Total amount (in Rs)
Karan	90	40	20	30	6600
Saroj	54	15	9	30	3150
Harish	55	30	5	20	2900
Jay	105	50	15	40	6300

Answer: B

According to the questions:

The ratio of the number of Rs 20 notes of Harish and Karan = a:b

Ratio of number of Rs 20 notes of Harish and Karan = 20:30 = 2:3

If Saroj had 'a*b' more notes of Rs 10

Total number of notes of Rs 10 = 6

Total amount of Saroj = 3150 + 60 = Rs 3210

Required answer = $(3210/6300) * 100 = 50.95\% \sim 51\%$

Hence the answer is option B.

2. Questions

For Saroj

Let the number of notes of Rs 50 = 5x

Number of notes of Rs 200 = 3x

Number of notes of Rs 20 = (3x+21)

$50*5x + 200*3x + 20*(3x+21) = 3150$

$250x + 600x + 60x + 420 = 3150$

$910x = 2730$

$x = 3$

The number of Rs 50 notes of Saroj = $5 * 3 = 15$

The number of Rs 200 notes of Saroj = $3 * 3 = 9$

The number of Rs 20 notes of Saroj = $9 + 21 = 30$

The ratio of number of Rs 50 notes of Karan, Saroj, Harish and Jay = 8:3:6:10 (40, 15, 30, 50) respectively

Let the number of Rs 20 notes of Harish = A

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$$30 \times 50 + 200 \times 5 + A \times 20 = 2900$$

$$2500 + 20A = 2900$$

$$20A = 400$$

$$A = 20$$

Let

The number of Rs 200 notes of Karan = x

The number of Rs 20 notes of Karan = 10 + x

$$10 + x + x + 40 = 90$$

$$2x = 40$$

$$x = 20$$

The number of Rs 200 notes of Karan = x = 20

The number of Rs 20 notes of Karan = 10 + x = 30

The total amount of Karan = $30 \times 20 + 40 \times 50 + 20 \times 200 = \text{Rs. } 6600$

Let

The number of Rs 200 notes of Jay = $50/10 \times 3 = 15$

The number of Rs 20 notes of Jay = 15 + 25 = 40

The total amount of Jay = $40 \times 20 + 50 \times 50 + 15 \times 200 = \text{Rs. } 6300$

Name	Total number of notes	Number of notes of Rs. 50	Number of notes of Rs. 200	Number of notes of Rs. 20	Total amount (in Rs)
Karan	90	40	20	30	6600
Saroj	54	15	9	30	3150
Harish	55	30	5	20	2900
Jay	105	50	15	40	6300

Answer: C

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I). The difference between the amount of Karan and Jay = $6600 - 6300 = \text{Rs } 300$ (true)

II). The sum of total number of Rs 200 notes of all four persons = $20 + 9 + 5 + 15 = 49$ (true)

III).

Total number of notes of Harish = 55

Total number of notes of Karan = 90

Required answer = $(55/90) \times 100 = 61.11\%$ (false)

Hence the answer is option C.

3. Questions

For Saroj

Let the number of notes of Rs 50 = $5x$

Number of notes of Rs 200 = $3x$

Number of notes of Rs 20 = $(3x+21)$

$$50*5x + 200*3x + 20*(3x+21) = 3150$$

$$250x + 600x + 60x + 420 = 3150$$

$$910x = 2730$$

$$x = 3$$

The number of Rs 50 notes of Saroj = $5 * 3 = 15$

The number of Rs 200 notes of Saroj = $3 * 3 = 9$

The number of Rs 20 notes of Saroj = $9 + 21 = 30$

The ratio of number of Rs 50 notes of Karan, Saroj, Harish and Jay = 8:3:6:10(40, 15, 30, 50) respectively

Let the number of Rs 20 notes of Harish = A

$$30*50 + 200*5 + A*20 = 2900$$

$$2500 + 20A = 2900$$

$$20A = 400$$

$$A = 20$$

Let

The number of Rs 200 notes of Karan = x

The number of Rs 20 notes of Karan = $10 + x$

$$10 + x + x + 40 = 90$$

$$2x = 40$$

$$x = 20$$

The number of Rs 200 notes of Karan = $x = 20$

The number of Rs 20 notes of Karan = $10 + x = 30$

The total amount of Karan = $30 * 20 + 40 * 50 + 20 * 200 = \text{Rs. } 6600$

Let

The number of Rs 200 notes of Jay = $50/10 * 3 = 15$

The number of Rs 20 notes of Jay = $15 + 25 = 40$

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The total amount of Jay = $40 * 20 + 50 * 50 + 15 * 200 = \text{Rs. } 6300$

Name	Total number of notes	Number of notes of Rs. 50	Number of notes of Rs. 200	Number of notes of Rs. 20	Total amount (in Rs)
Karan	90	40	20	30	6600
Saroj	54	15	9	30	3150
Harish	55	30	5	20	2900
Jay	105	50	15	40	6300

Answer: D

Q I:

Total number of notes of Rs 50 of Harish and Jay = $30+50 = 80$

Total number of total notes of Rs 20 of Saroj and Harish = $30+30 = 60$

Required answer = $80 - 60 = 20$

Q II:

Average number of total notes of Jay and Harish = $(55+105)/2 = 160/2 = 80$

Total number of notes of Rs 200 of all four = $20+9+5+15 = 49$

Required answer = $80 - 49 = 31$

Q III:

20% of the difference between the amount of Saroj and Harish

Required answer = $(3150 - 2900) * 20\% = 50$

$QI < QII < QIII$

Hence the answer is option D

4. Questions

For Saroj

Let the number of notes of Rs 50 = $5x$

Number of notes of Rs 200 = $3x$

Number of notes of Rs 20 = $(3x+21)$

$50*5x + 200*3x + 20*(3x+21) = 3150$

$250x + 600x + 60x + 420 = 3150$

$910x = 2730$

$x = 3$

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The number of Rs 50 notes of Saroj = $5 * 3 = 15$

The number of Rs 200 notes of Saroj = $3 * 3 = 9$

The number of Rs 20 notes of Saroj = $9 + 21 = 30$

The ratio of number of Rs 50 notes of Karan, Saroj, Harish and Jay = 8:3:6:10 (40, 15, 30, 50) respectively

Let the number of Rs 20 notes of Harish = A

$$30 * 50 + 200 * 5 + A * 20 = 2900$$

$$2500 + 20A = 2900$$

$$20A = 400$$

$$A = 20$$

Let

The number of Rs 200 notes of Karan = x

The number of Rs 20 notes of Karan = $10 + x$

$$10 + x + x + 40 = 90$$

$$2x = 40$$

$$x = 20$$

The number of Rs 200 notes of Karan = $x = 20$

The number of Rs 20 notes of Karan = $10 + x = 30$

The total amount of Karan = $30 * 20 + 40 * 50 + 20 * 200 = \text{Rs. } 6600$

Let

The number of Rs 200 notes of Jay = $50/10 * 3 = 15$

The number of Rs 20 notes of Jay = $15 + 25 = 40$

The total amount of Jay = $40 * 20 + 50 * 50 + 15 * 200 = \text{Rs. } 6300$

Name	Total number of notes	Number of notes of Rs. 50	Number of notes of Rs. 200	Number of notes of Rs. 20	Total amount (in Rs)
Karan	90	40	20	30	6600
Saroj	54	15	9	30	3150
Harish	55	30	5	20	2900
Jay	105	50	15	40	6300

Answer: C

Statement I

Total amount of Deep = Rs 6900

Total number of notes of Rs 20 = 45

This statement alone is not sufficient to answer.

Statement II

Difference between the number of notes of Rs 20 and 200 of Deep = 30

Number of notes of Rs 50 of Deep = $50 \times 120\% = 60$

This statement alone is not sufficient to answer.

Taking both statements together

Number of notes of Rs 50 of Deep = 60

Total number of notes of Rs 20 = 45

Difference between the number of notes of Rs 20 and 200 of Deep = 30

Number of notes of Rs 200 = 15 or 75 (75 is eliminated because $75 \times 200 = 15000$ is more than 6900)

Number of notes of Rs 200 = 15

Amount = $50 \times 60 + 20 \times 45 + 15 \times 200 = \text{Rs } 6900$

Total number of notes of Deep = $60 + 45 + 15 = 120$

These two statements together are sufficient to answer.

Hence the answer is option C.

5. Questions

For Saroj

Let the number of notes of Rs 50 = $5x$

Number of notes of Rs 200 = $3x$

Number of notes of Rs 20 = $(3x + 21)$

$50 \times 5x + 200 \times 3x + 20 \times (3x + 21) = 3150$

$250x + 600x + 60x + 420 = 3150$

$910x = 2730$

$x = 3$

The number of Rs 50 notes of Saroj = $5 \times 3 = 15$

The number of Rs 200 notes of Saroj = $3 \times 3 = 9$

The number of Rs 20 notes of Saroj = $9 + 21 = 30$

The ratio of number of Rs 50 notes of Karan, Saroj, Harish and Jay = $8:3:6:10$ (40, 15, 30, 50) respectively

Let the number of Rs 20 notes of Harish = A

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$$30 \times 50 + 200 \times 5 + A \times 20 = 2900$$

$$2500 + 20A = 2900$$

$$20A = 400$$

$$A = 20$$

Let

The number of Rs 200 notes of Karan = x

The number of Rs 20 notes of Karan = 10 + x

$$10 + x + x + 40 = 90$$

$$2x = 40$$

$$x = 20$$

The number of Rs 200 notes of Karan = x = 20

The number of Rs 20 notes of Karan = 10 + x = 30

The total amount of Karan = $30 \times 20 + 40 \times 50 + 20 \times 200 = \text{Rs. } 6600$

Let

The number of Rs 200 notes of Jay = $50/10 \times 3 = 15$

The number of Rs 20 notes of Jay = $15 + 25 = 40$

The total amount of Jay = $40 \times 20 + 50 \times 50 + 15 \times 200 = \text{Rs. } 6300$

Name	Total number of notes	Number of notes of Rs. 50	Number of notes of Rs. 200	Number of notes of Rs. 20	Total amount (in Rs)
Karan	90	40	20	30	6600
Saroj	54	15	9	30	3150
Harish	55	30	5	20	2900
Jay	105	50	15	40	6300

Answer: A

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Total number of notes of Mahesh is 8 more than average number of total notes of Karan and Saroj.

$$\text{Total number of notes of Mahesh} = 8 + (90 + 54)/2 = 80$$

$$\text{Number of notes of Rs 50} = 25$$

$$\text{Total number of notes of Rs 20 and Rs 200} = 80 - 25 = 55$$

He has highest number of notes of Rs 200

$$\text{If number of notes of Rs 200} = 26$$

Then number of notes of Rs 20 = $55 - 26 = 29$

(not possible)

If number of notes of Rs 200 = 27

Then number of notes of Rs 20 = $55 - 27 = 28$

(not possible)

If number of notes of Rs 200 = 28

Then number of notes of Rs 20 = $55 - 28 = 27$

(possible)

Required answer = 27

Hence the answer is option A.

6. Questions

Radius of circle = $(6x + y - 1)$ cm

Perimeter of circle = 132 cm

$$2 \times \frac{22}{7} \times (6x + y - 1) = 132$$

$$6x + y = 22 \dots\dots\dots(1)$$

Radius of sphere = $\frac{3(x + y)}{2}$ cm

Volume of sphere = 4851 cm^3

$$\left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times \left[\frac{3(x + y)}{2}\right]^3 = 4851$$

$$\left[\frac{(x + y)}{2}\right]^3 = 49 \times \frac{7}{8}$$

$$\frac{(x + y)}{2} = \frac{7}{2}$$

$$x + y = 7 \dots\dots\dots(2)$$

Solving both equations

We get

$$x = 3 \text{ and } y = 4$$

Slant height of cone = 25 cm

$$\text{Height of cone} = \sqrt{(25^2 - 7^2)} = 24 \text{ cm}$$

Height of cylinder is x cm more than that of cone.

$$\text{Height of cylinder} = 24 + 3 = 27 \text{ cm}$$

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Shape	Radius (in cm)
Cylinder	14
Cone	7
Circle	21
Sphere	10.5

Answer: D

Radius of circle = 21 cm

Height of cylinder = 27 cm

Required answer = $21 \times 100 / 27 = 77.78\%$ (false)

Curved surface area of cone = $22/7 \times 7 \times 25 = 550 \text{ cm}^2$ (false)

Average of radius of cone and circle = $(7 + 21) / 2 = 14 \text{ cm}$ (false)

All statements are false.

Hence the answer is option D.

7. Questions

Radius of circle = $(6x + y - 1) \text{ cm}$

Perimeter of circle = 132 cm

$$2 \times 22/7 \times (6x + y - 1) = 132$$

$$6x + y = 22 \dots\dots\dots(1)$$

Radius of sphere = $3(x + y) / 2 \text{ cm}$

Volume of sphere = 4851 cm^3

$$(4/3) \times (22/7) \times [3(x + y) / 2]^3 = 4851$$

$$[(x + y) / 2]^3 = 49 \times 7/8$$

$$(x + y) / 2 = 7/2$$

$$x + y = 7 \dots\dots\dots(2)$$

Solving both equations

We get

$$x = 3 \text{ and } y = 4$$

Slant height of cone = 25 cm

$$\text{Height of cone} = \sqrt{(25^2 - 7^2)} = 24 \text{ cm}$$

Height of cylinder is x cm more than that of cone.

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Height of cylinder = $24+3 = 27$ cm

Shape	Radius (in cm)
Cylinder	14
Cone	7
Circle	21
Sphere	10.5

Answer: D

Q I:

$$\text{Area of circle} = \frac{22}{7} \times 21 \times 21 = 1386 \text{ cm}^2$$

$$\text{Area of base of cylinder} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

$$\text{Required answer} = 1386 - 616 = 770 \text{ cm}^2$$

Q II:

$$\text{Total surface area of sphere} = 4 \times \frac{22}{7} \times 10.5 \times 10.5 = 1386 \text{ cm}^2$$

Q III:

$$\text{Curved surface area of cylinder} = 2 \times \frac{22}{7} \times 14 \times 27 = 2376 \text{ cm}^2$$

QI < QII < QIII

Hence the answer is option D.

8. Questions

$$\text{Radius of circle} = (6x+y-1) \text{ cm}$$

$$\text{Perimeter of circle} = 132 \text{ cm}$$

$$2 \times \frac{22}{7} \times (6x+y-1) = 132$$

$$6x+y = 22 \dots\dots\dots(1)$$

$$\text{Radius of sphere} = 3(x+y)/2 \text{ cm}$$

$$\text{Volume of sphere} = 4851 \text{ cm}^3$$

$$\left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times \left[\frac{3(x+y)}{2}\right]^3 = 4851$$

$$\left[\frac{(x+y)}{2}\right]^3 = 49 \times \frac{7}{8}$$

$$(x+y)/2 = 7/2$$

$$x+y = 7 \dots\dots\dots(2)$$

Solving both equations

@QuizByMickey

We get

$$x = 3 \text{ and } y = 4$$

Slant height of cone = 25 cm

$$\text{Height of cone} = \sqrt{(25^2 - 7^2)} = 24 \text{ cm}$$

Height of cylinder is x cm more than that of cone.

$$\text{Height of cylinder} = 24 + 3 = 27 \text{ cm}$$

Shape	Radius (in cm)
Cylinder	14
Cone	7
Circle	21
Sphere	10.5

Answer: A

$$\text{Height of cuboid} = 14 + 6 = 20 \text{ cm}$$

$$\text{Base of cuboid} = 24 - 10 = 14 \text{ cm}$$

$$\text{Length of cuboid} = 3x = 9 \text{ cm}$$

$$\text{Total surface area of cuboid} = 2(20 \times 14 + 14 \times 9 + 9 \times 20) = 1172 \text{ cm}^2$$

Hence the answer is option A.

9. Questions

$$\text{Radius of circle} = (6x + y - 1) \text{ cm}$$

$$\text{Perimeter of circle} = 132 \text{ cm}$$

$$2 \times \frac{22}{7} \times (6x + y - 1) = 132$$

$$6x + y = 22 \dots\dots\dots(1)$$

$$\text{Radius of sphere} = \frac{3(x + y)}{2} \text{ cm}$$

$$\text{Volume of sphere} = 4851 \text{ cm}^3$$

$$\left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times \left[\frac{3(x + y)}{2}\right]^3 = 4851$$

$$\left[\frac{(x + y)}{2}\right]^3 = 49 \times \frac{7}{8}$$

$$\frac{(x + y)}{2} = \frac{7}{2}$$

$$x + y = 7 \dots\dots\dots(2)$$

Solving both equations

We get

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$x = 3$ and $y = 4$

Slant height of cone = 25 cm

Height of cone = $\sqrt{(25^2 - 7^2)} = 24$ cm

Height of cylinder is x cm more than that of cone.

Height of cylinder = $24 + 3 = 27$ cm

Shape	Radius (in cm)
Cylinder	14
Cone	7
Circle	21
Sphere	10.5

Answer: B

The given sphere is melted and formed into $(30y+n+1)$ number of small spherical marbles whose radius is 1.5 cm.

$$(4/3) * \pi * (10.5)^3 = (30*4+n+1) * (4/3) * \pi * (1.5)^3$$

$$n = 222$$

$$\text{Required answer} = 222/3 = 74$$

Hence the answer is option B.

10. Questions

Radius of circle = $(6x+y-1)$ cm

Perimeter of circle = 132 cm

$$2 * 22/7 * (6x+y-1) = 132$$

$$6x+y = 22 \dots\dots\dots(1)$$

Radius of sphere = $3(x+y)/2$ cm

Volume of sphere = 4851 cm³

$$(4/3) * (22/7) * [3(x+y)/2]^3 = 4851$$

$$[(x+y)/2]^3 = 49 * 7/8$$

$$(x+y)/2 = 7/2$$

$$x+y = 7 \dots\dots\dots(2)$$

Solving both equations

We get

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$$x = 3 \text{ and } y = 4$$

Slant height of cone = 25 cm

$$\text{Height of cone} = \sqrt{(25^2 - 7^2)} = 24 \text{ cm}$$

Height of cylinder is x cm more than that of cone.

$$\text{Height of cylinder} = 24 + 3 = 27 \text{ cm}$$

Shape	Radius (in cm)
Cylinder	14
Cone	7
Circle	21
Sphere	10.5

Answer: E

Breadth of a rectangle = 25 cm

$$\text{Length of rectangle} = 21 + 9 = 30 \text{ cm}$$

Perimeter of a square is same as the perimeter of circle.

Let the side of square = A cm

$$4A = 132$$

$$A = 33 \text{ cm}$$

$$\text{Area of rectangle} = 25 \times 30 = 750 \text{ cm}^2$$

$$\text{Area of square} = 33 \times 33 = 1089 \text{ cm}^2$$

$$\text{Required answer} = 1089 - 750 = 339 \text{ cm}^2$$

Hence the answer is option E.

11. Questions

Total number of people who use both Puma and Addidas = 38

Total number of people using Puma = 95

Ratio of the number of people who use only Puma and only Addidas = 11:40

Number of people who use only Puma = 11x

Number of people who use Addidas only = 40x

Total number of people using Gucci brand is 42.5% more than the number of people who use only Addidas.

$$\text{Total number of people using Gucci brand} = 40x \times 142.5\% = 57x$$

The number of people who use both Gucci and Addidas but not Puma is 100 less than the total number of

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people using Gucci.

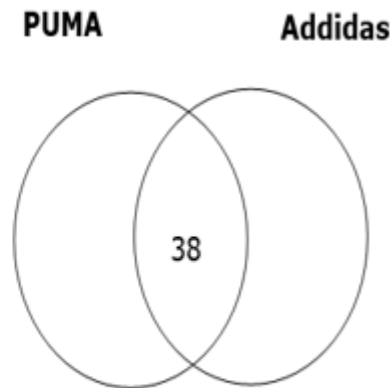
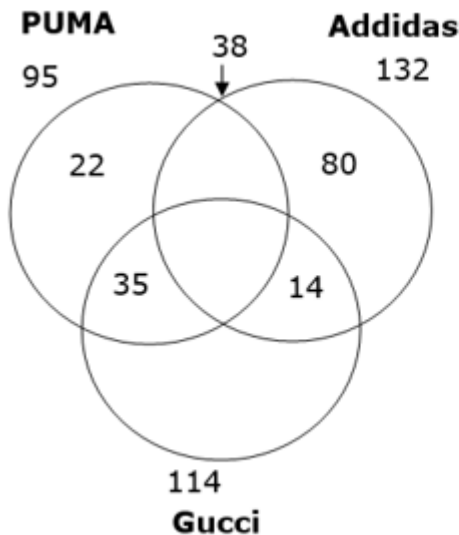
The number of people who use both Gucci and Addidas but not Puma = $(57x-100)$

Total number of people using Addidas = 132

$$38 + 40x + 57x - 100 = 132$$

$$97x = 194$$

$$x = 2$$



Answer: E

We cannot find the required answer for the given question.

Hence the answer is option E.

12. Questions

Total number of people who use both Puma and Addidas = 38

Total number of people using Puma = 95

Ratio of the number of people who use only Puma and only Addidas = 11:40

Number of people who use only Puma = $11x$

Number of people who use Addidas only = $40x$

Total number of people using Gucci brand is 42.5% more than the number of people who use only Addidas.

Total number of people using Gucci brand = $40x * 142.5\% = 57x$

The number of people who use both Gucci and Addidas but not Puma is 100 less than the total number of people using Gucci.

The number of people who use both Gucci and Addidas but not Puma = $(57x-100)$

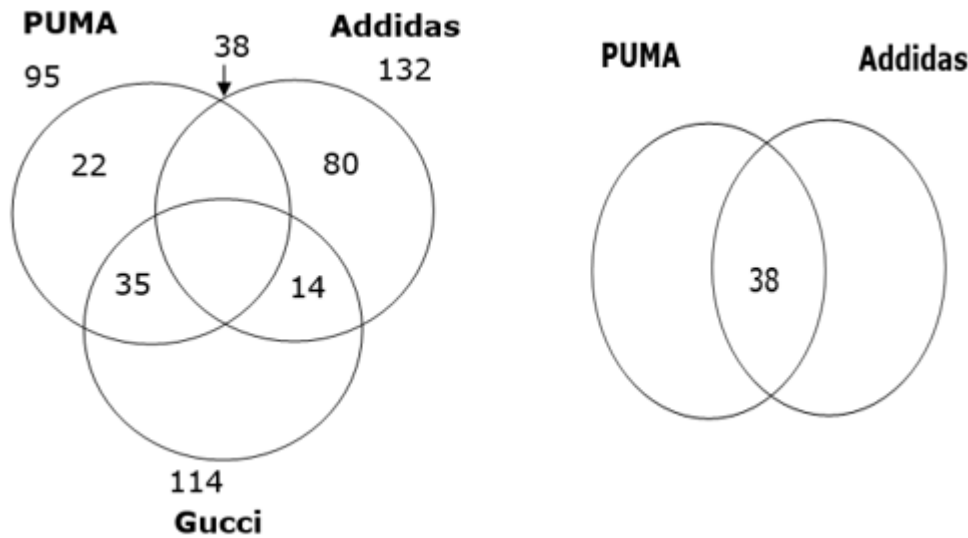
Total number of people using Addidas = 132

$$38 + 40x + 57x - 100 = 132$$

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$$97x = 194$$

$$x = 2$$



Answer: A

Number of people who use exactly three brands = 10

Total number of people using exactly two brands = $(38-10)+14+35 = 77$

Total number of sport persons = $114+28+22+80 = 244$

Required answer = $77*100/244 = 31.55\% \sim 32\%$

Hence the answer is option A.

13. Questions

Total number of people who use both Puma and Addidas = 38

Total number of people using Puma = 95

Ratio of the number of people who use only Puma and only Addidas = 11:40

Number of people who use only Puma = $11x$

Number of people who use Addidas only = $40x$

Total number of people using Gucci brand is 42.5% more than the number of people who use only Addidas.

Total number of people using Gucci brand = $40x*142.5\% = 57x$

The number of people who use both Gucci and Addidas but not Puma is 100 less than the total number of people using Gucci.

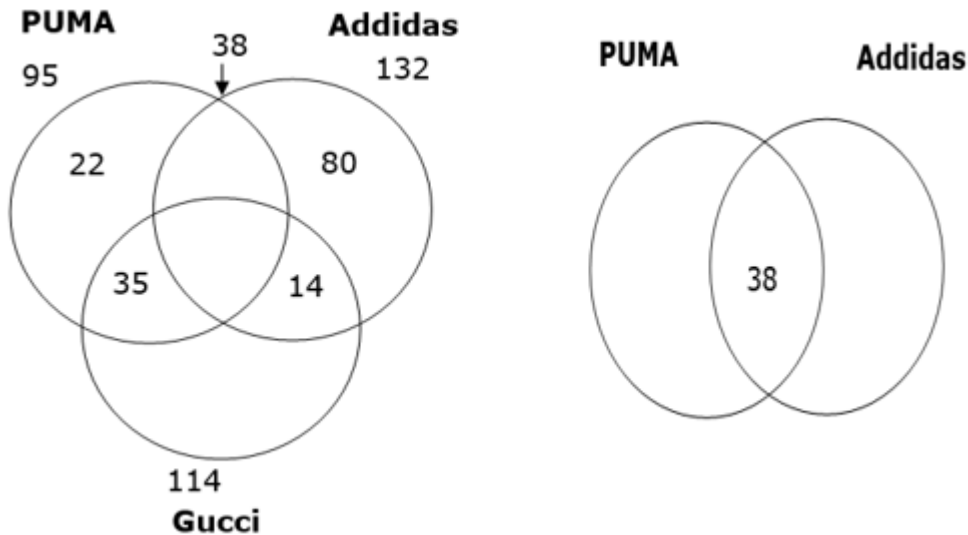
The number of people who use both Gucci and Addidas but not Puma = $(57x-100)$

Total number of people using Addidas = 132

$$38+40x+57x-100=132$$

$$97x = 194$$

$$x = 2$$



Answer: A

I). Required ratio = $14:35 = 2:5$ (True)

II). Required percentage = $95 \times 100 / 114 = 83.33\%$ (False)

III). Required average = $(22 + 80) / 2 = 51$ (False)

Hence the answer is option A.

14. Questions

Total number of people who use both Puma and Addidas = 38

Total number of people using Puma = 95

Ratio of the number of people who use only Puma and only Addidas = 11:40

Number of people who use only Puma = $11x$

Number of people who use Addidas only = $40x$

Total number of people using Gucci brand is 42.5% more than the number of people who use only Addidas.

Total number of people using Gucci brand = $40x \times 142.5\% = 57x$

The number of people who use both Gucci and Addidas but not Puma is 100 less than the total number of people using Gucci.

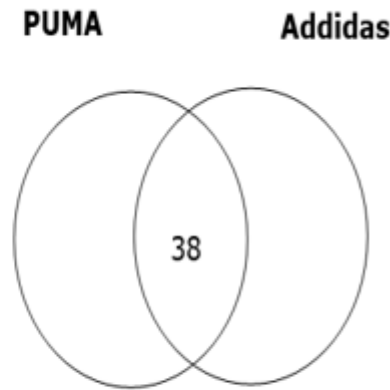
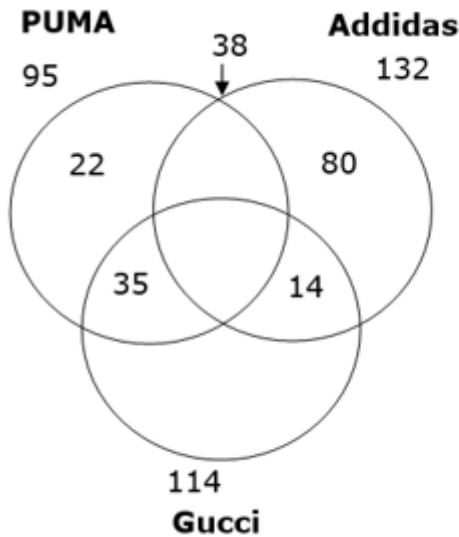
The number of people who use both Gucci and Addidas but not Puma = $(57x - 100)$

Total number of people using Addidas = 132

$$38 + 40x + 57x - 100 = 132$$

$$97x = 194$$

$$x = 2$$



Answer: A

Q I:

Number of people who use both puma and Addidas but not Gucci = $2 \times 14 = 28$

Total number of people who use exactly one brand = $22 + 80 + (114 - 35 - 14 - 10) = 157$

Q II:

Average of total number of people using Puma, Addidas and Gucci = $(95 + 132 + 114)/3 = 341/3 = 113.66$

QI > QII

Hence the answer is option A.

15. Questions

Total number of people who use both Puma and Addidas = 38

Total number of people using Puma = 95

Ratio of the number of people who use only Puma and only Addidas = 11:40

Number of people who use only Puma = $11x$

Number of people who use Addidas only = $40x$

Total number of people using Gucci brand is 42.5% more than the number of people who use only Addidas.

Total number of people using Gucci brand = $40x \times 142.5\% = 57x$

The number of people who use both Gucci and Addidas but not Puma is 100 less than the total number of people using Gucci.

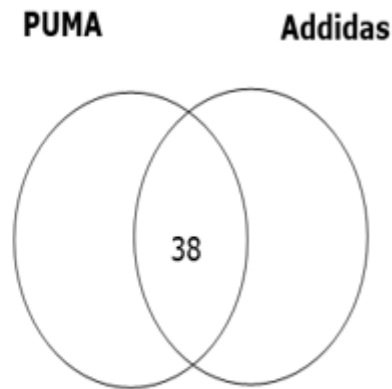
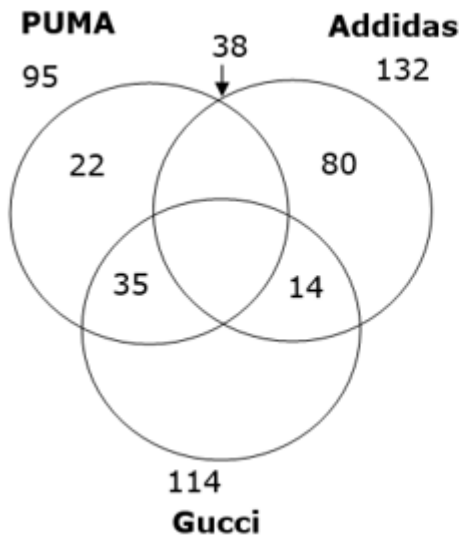
The number of people who use both Gucci and Addidas but not Puma = $(57x - 100)$

Total number of people using Addidas = 132

$38 + 40x + 57x - 100 = 132$

$$97x = 194$$

$$x = 2$$



Answer: C

$$'m' = 95 - 80 = 15$$

$$'n' = 22 + 14 = 36$$

$$\text{Total number of people using exactly all three brands} = 36 - 15 = 21$$

$$\text{Total number of sport persons} = 114 + 22 + 80 + (38 - 21) = 233$$

Hence the answer is option C.

16. Questions

Answer: D

$$B = 36^2 + 200$$

$$B = 1496$$

$$A = 50\% \text{ of } B + 104$$

$$A = 50\% \text{ of } 1496 + 104$$

$$A = 852$$

Series:

1500	1492	1556	1340	1852	852
-8	+64	-216	+512	-1000	
2^3	4^3	6^3	8^3	10^3	

Wrong term = 1496

Correct term = 1492

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Hence the answer is option D.

17. Questions

Answer: B

$$(3x+2y) = 2440$$

$$(2x-y) = 460$$

Solving both equations

We get

$$x = 480 \text{ and } y = 500$$

Series: 4000, 800, 2400, 480, _____, 288

4000	800	2400	480	1440	288
÷5	*3	÷5	*3	÷5	

$$\text{Missing term} = 480 * 3 = 1440$$

$$\text{First term of new series} = 60 + 1440 = 1500$$

New series:

1500	30	900	180,
÷5	*3	÷5	

$$\text{Required answer} = 900$$

Hence the answer is option B.

18. Questions

Answer: C

$$A/B = 100$$

$$A+10B = 275$$

Solving both equations

We get

$$A = 250 \text{ and } B = 2.5$$

Series 1:

10000	250	12.5	1.25	0.25
÷40	÷20	÷10	÷5	

$$\text{Missing term} = 1.25/5 = 0.25$$

Series 2: 10A, D, _____, C, E

2500	62.5	3.125
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$\div 40$

$\div 20$

Series 2: 2500, 62.5, 3.125, C, E

Required answer = $0.25 + 3.125 = 3.375$

Hence the answer is option C.

19. Questions

Answer: A

Series 1:

1250	1183	1112	1039	960
-67	-71	-73	-79	

Wrong term = 1000

Correct term = $1039 - 79 = 960$

New series: 1000, 933, 862, 789, 710

1000	933	862	789	710
-67	-71	-73	-79	

Required answer = $(789 + 710)/2 = 749.5$

Hence the answer is option A.

20. Questions

Answer: B

$$(X+Y) = 920$$

$$(Y-Z) = 230$$

$$(X-Z) = 46$$

Solving all three equations

We get

$$x = 368, y = 552 \text{ and } z = 322$$

Series 1: Z, X, 437, Y, 710, 966

Series 1: 322, 368, 437, 552, 710, 966

322	368	437	552	713	966
+46	+69	+115	+161	+253	
23×2	23×3	23×5	23×7	23×11	

Wrong term = 710

Correct term = $552 + 161 = 713$

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Required answer = $966 - 713 = 253$

Hence the answer is option B.

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